

Introduction to C Programming

Module 14.5: Practice Day 01

(Codeforces Links and Practice Questions)

Topics:

1. Nested Loop
2. Pattern Printing
3. Functions

Codeforces Problem Links:

1. **P. Shape1**
   Solution link: P Shape1 - YouTube

2. **T. Shape2**
   Solution link: T Shape 2

3. **W. Shape3**
   Solution link: W Shape 3

4. **H. Sorting**
   Solution link: H Sorting

5. **N. Shift Zeros**
   Solution link: N Shift Zeros

6. **G. Max and MIN**
   Solution link: G Max and Min

7. **W. Mathematical Expression**
   Solution link: W Mathematical Expression

[Scroll down to see more practice problems]

[***Try this also, if you haven't done it before***
https://docs.google.com/document/d/1KyUB7JrpN3tinY1C1epwbG4nfK5uAuGZG_da36ksnl0/edit?usp=sharing]



Question: Take a positive integer N as input and print the pattern shown in the sample input output.

Sample Input	Sample Output
5	* ** *** **** *****

7	 * ** *** **** ***** ****** *******

Practice Problem 2

Question: Take a positive integer N as input and print the pattern shown in the sample input output.

Sample Input	Sample Output
5	********* *******

	***** *** *
8	*************** ************* *********** ********* ******* ***** *** *



Question: Make a function named **char_to_ascii()** which will take a character as parameter and it will convert that character to ascii value and return through that function. So, the return type will be int. Now in the main function take a character input and call that function to get the integer value of it and print that in the main function.

Bonus: You can also try this with 4 different types of function use cases.

Sample Input	Sample Output
a	97
A	65
0	48



Question: Make a function named **small_to_capital()** which will take a small alphabet as a parameter (which is a character) and it will convert that small alphabet to capital alphabet and return through that function. So, the return type will be char. Now in the main function take a character input which will be small alphabets and call that function to get the capital alphabet and print that in the main function.

Bonus: You can also try this with 4 different types of function use cases.

Sample Input	Sample Output
a	A
z	Z
q	Q



Question: Make a function named **capital_to_small()** which will take a capital alphabet as a parameter (which is a character) and it will convert that capital alphabet to small alphabet and return through that function. So, the return type will be char. Now in the main function take a character input which will be capital alphabets and call that function to get the small alphabet and print that in the main function.

Bonus: You can also try this with 4 different types of function use cases.

Sample Input	Sample Output
A	a
Y	y
Q	q